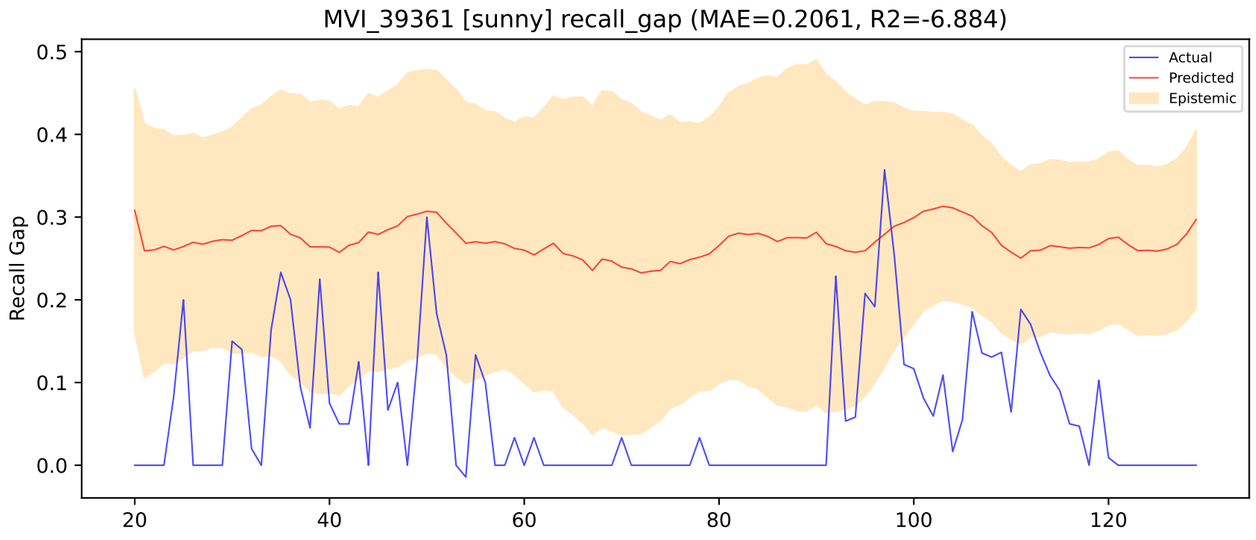
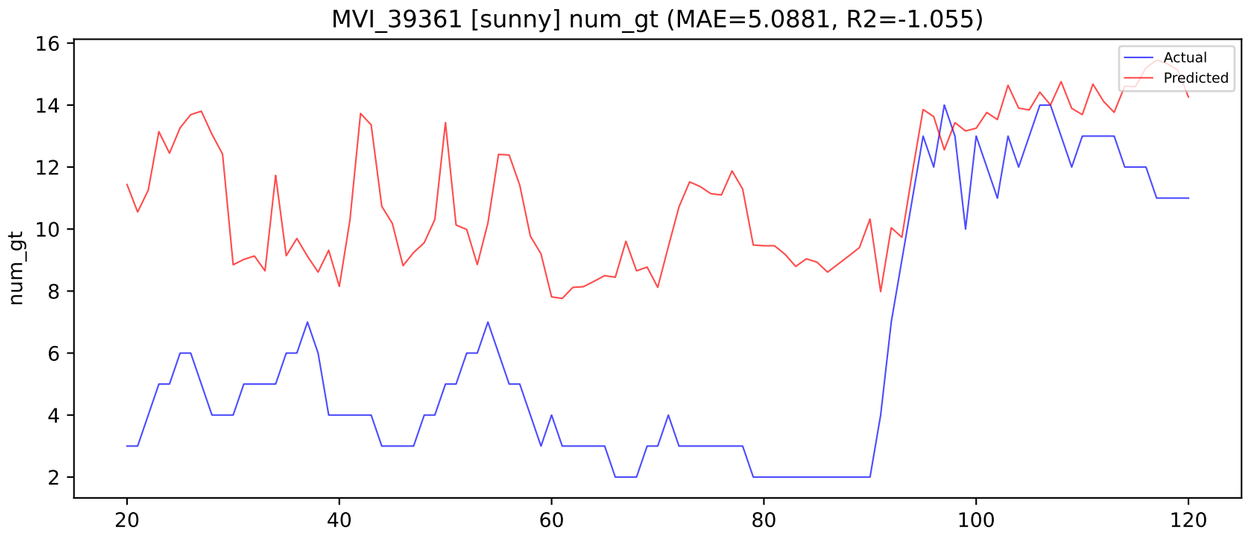
LSTM_medium (23 feat: 14 image + 9 medium detection) -- Top 2 Sequences per Weather

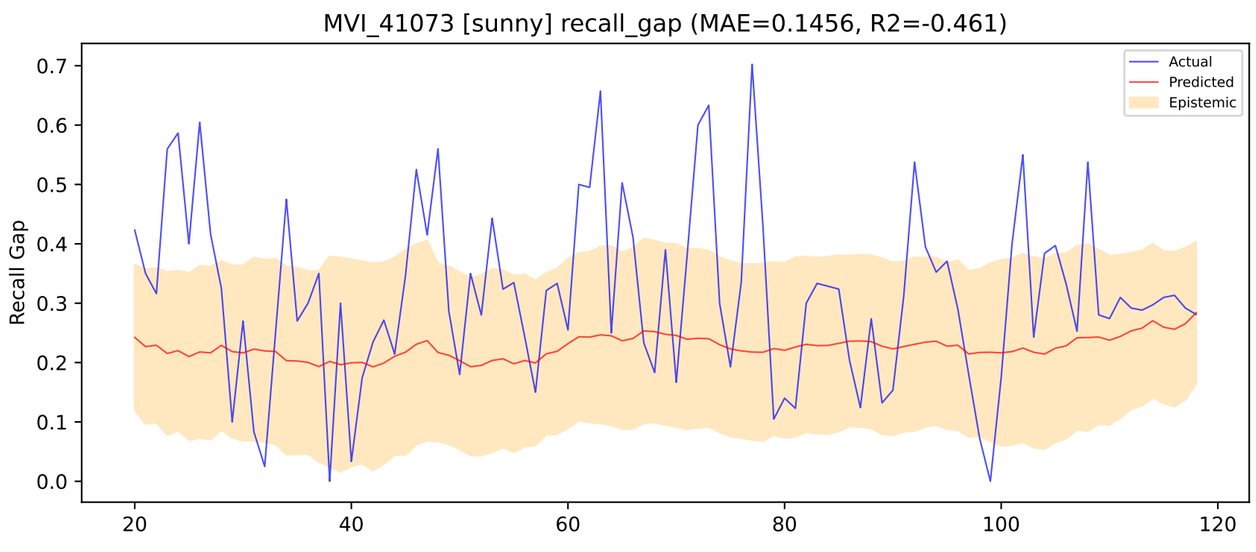
MVI_39361 [sunny] recall_gap (MAE=0.2061, R2=-6.884)



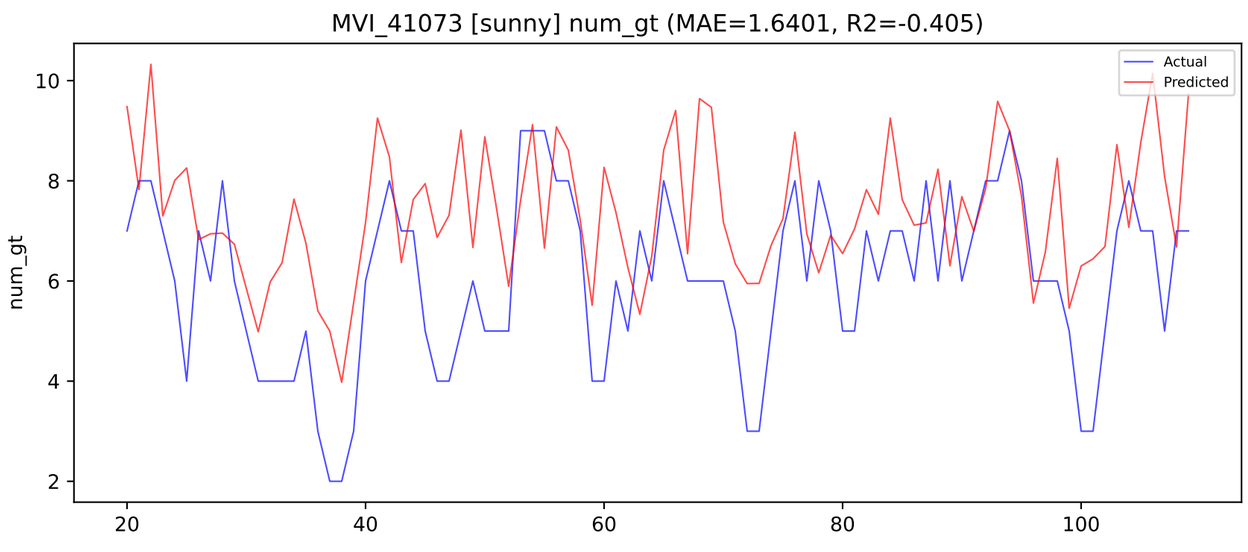
MVI_39361 [sunny] num_gt (MAE=5.0881, R2=-1.055)



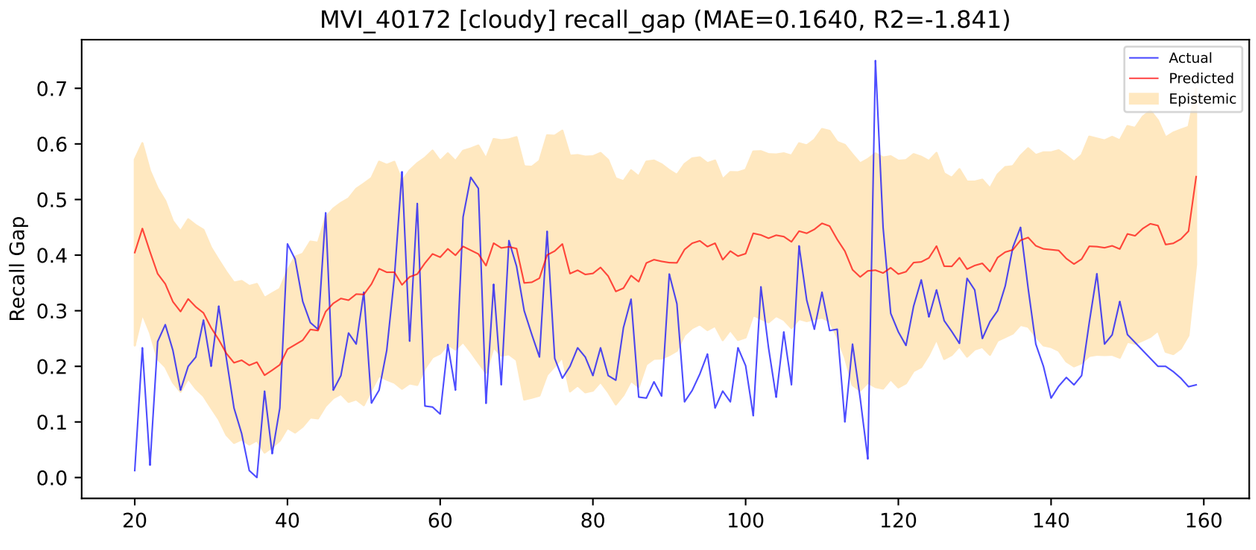
MVI_41073 [sunny] recall_gap (MAE=0.1456, R2=-0.461)



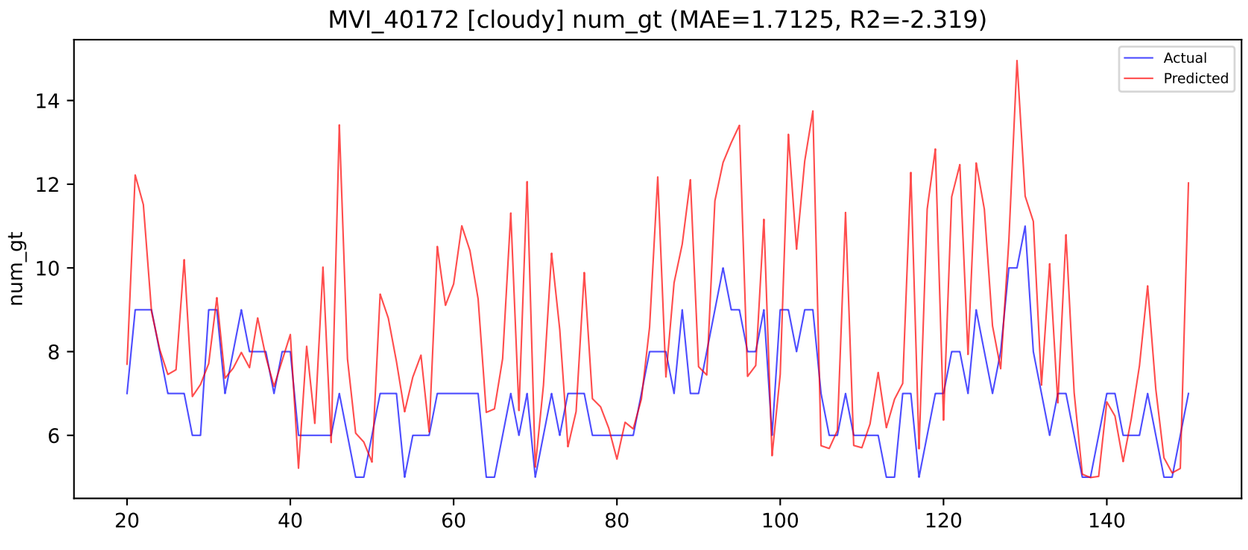
MVI_41073 [sunny] num_gt (MAE=1.6401, R2=-0.405)



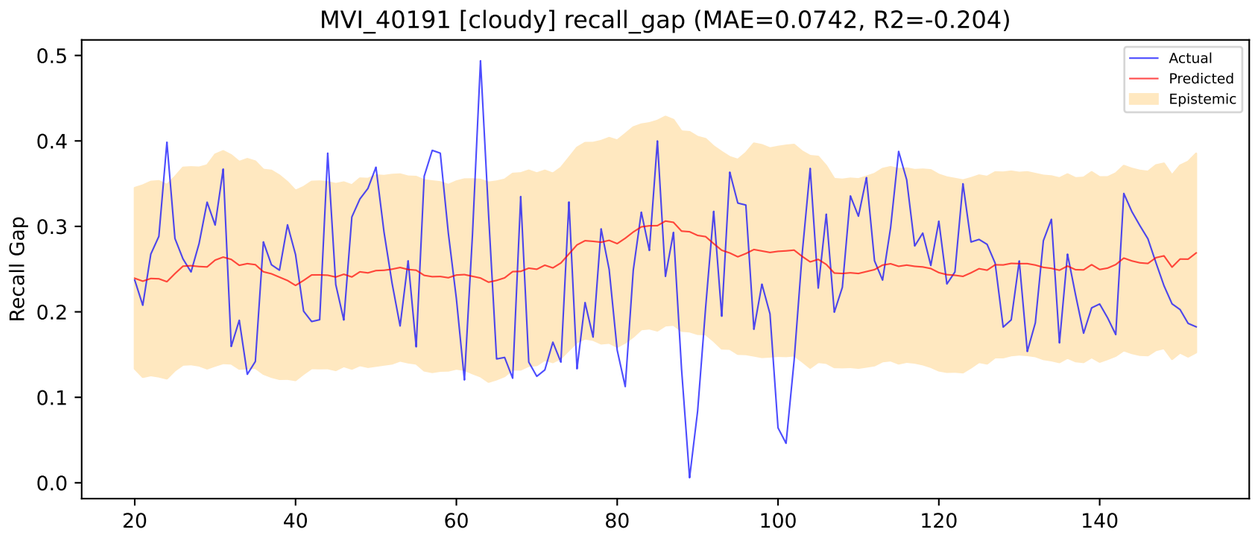
MVI_40172 [cloudy] recall_gap (MAE=0.1640, R2=-1.841)



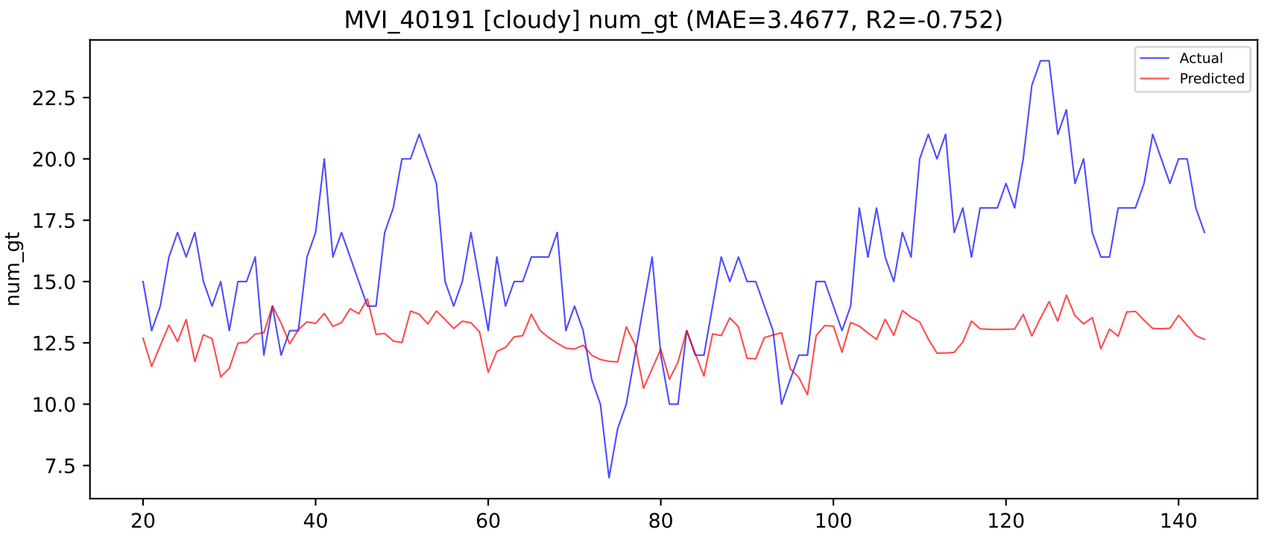
MVI_40172 [cloudy] num_gt (MAE=1.7125, R2=-2.319)



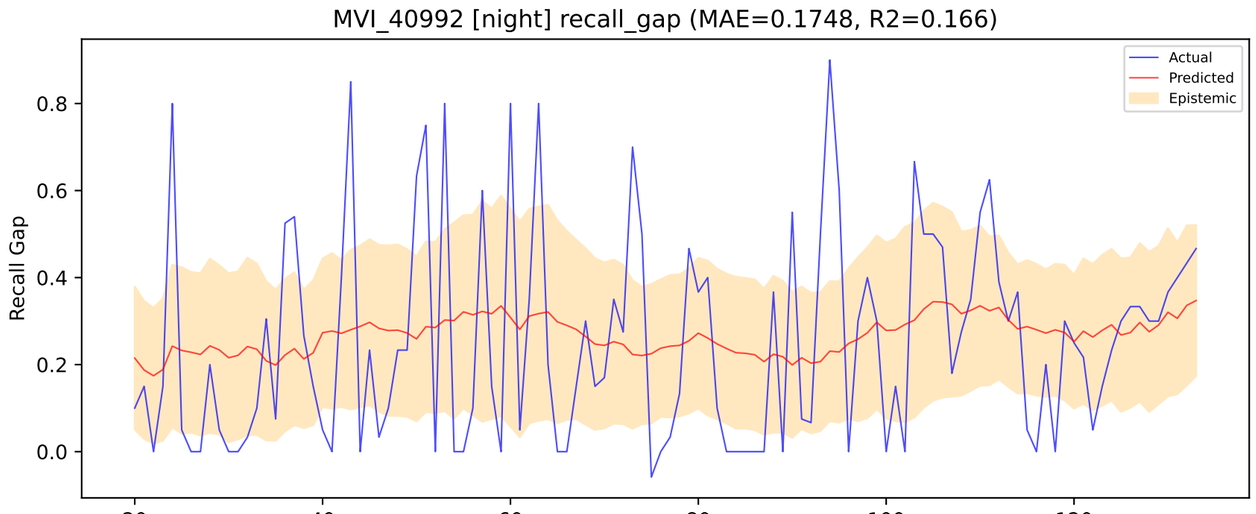
MVI_40191 [cloudy] recall_gap (MAE=0.0742, R2=-0.204)



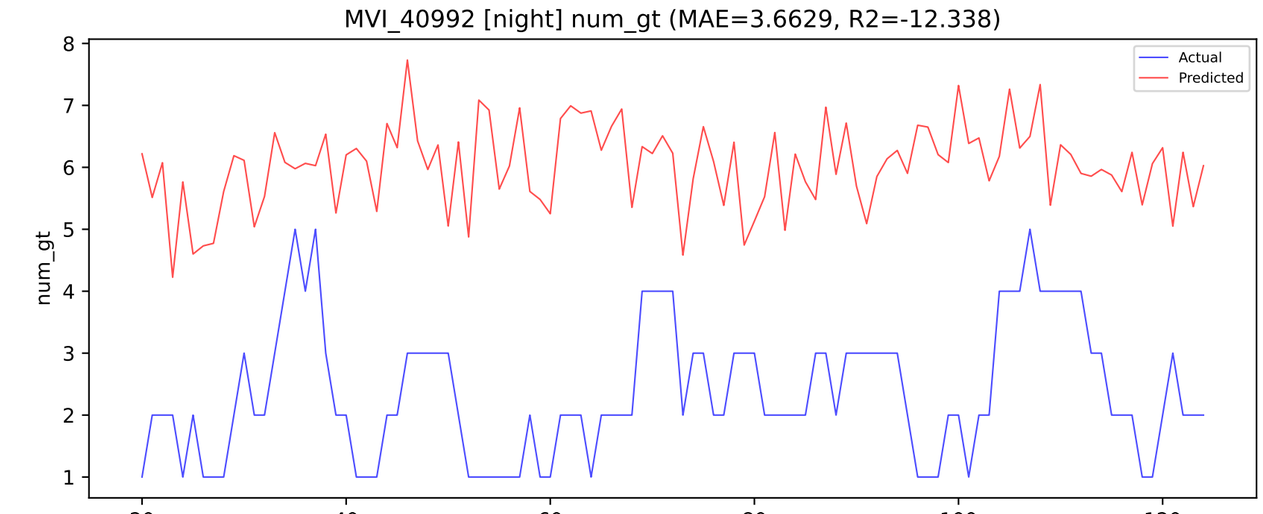
MVI_40191 [cloudy] num_gt (MAE=3.4677, R2=-0.752)



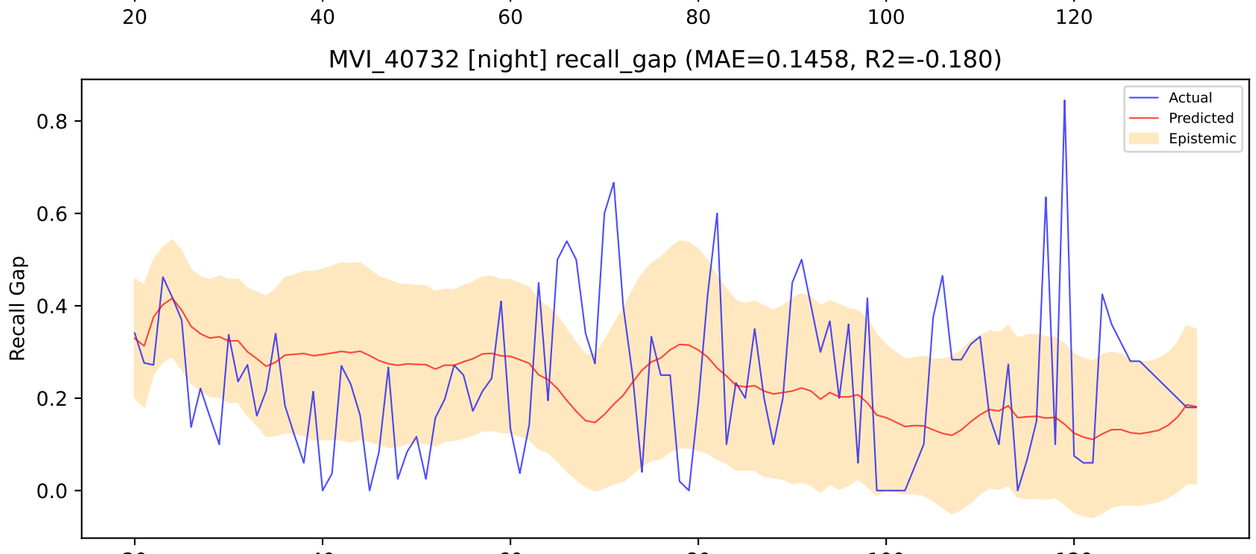
MVI_40992 [night] recall_gap (MAE=0.1748, R2=0.166)



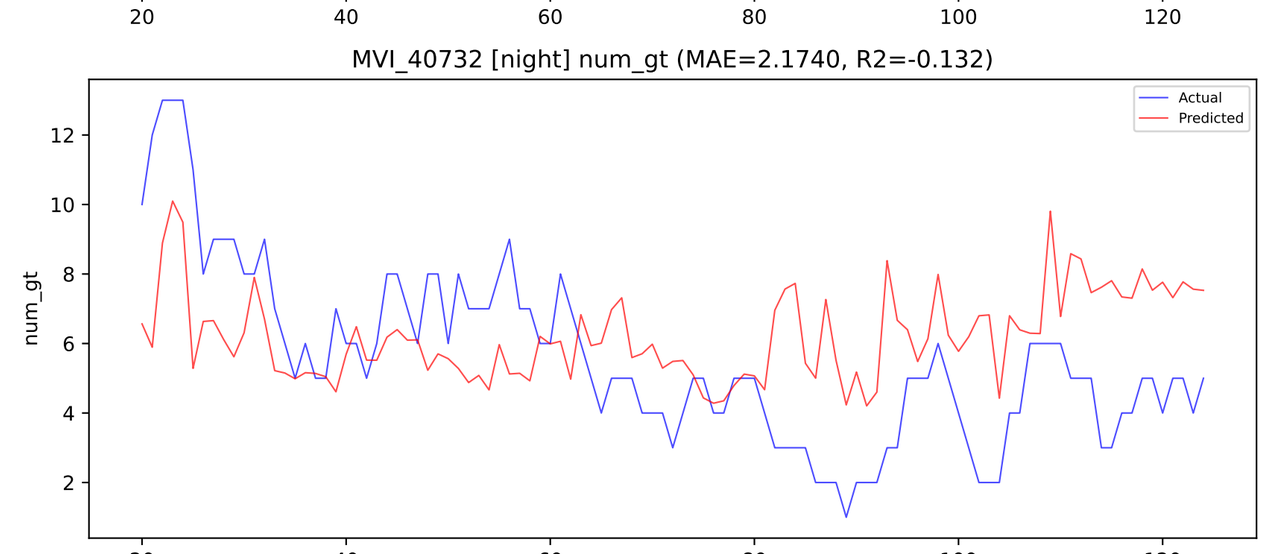
MVI_40992 [night] num_gt (MAE=3.6629, R2=-12.338)



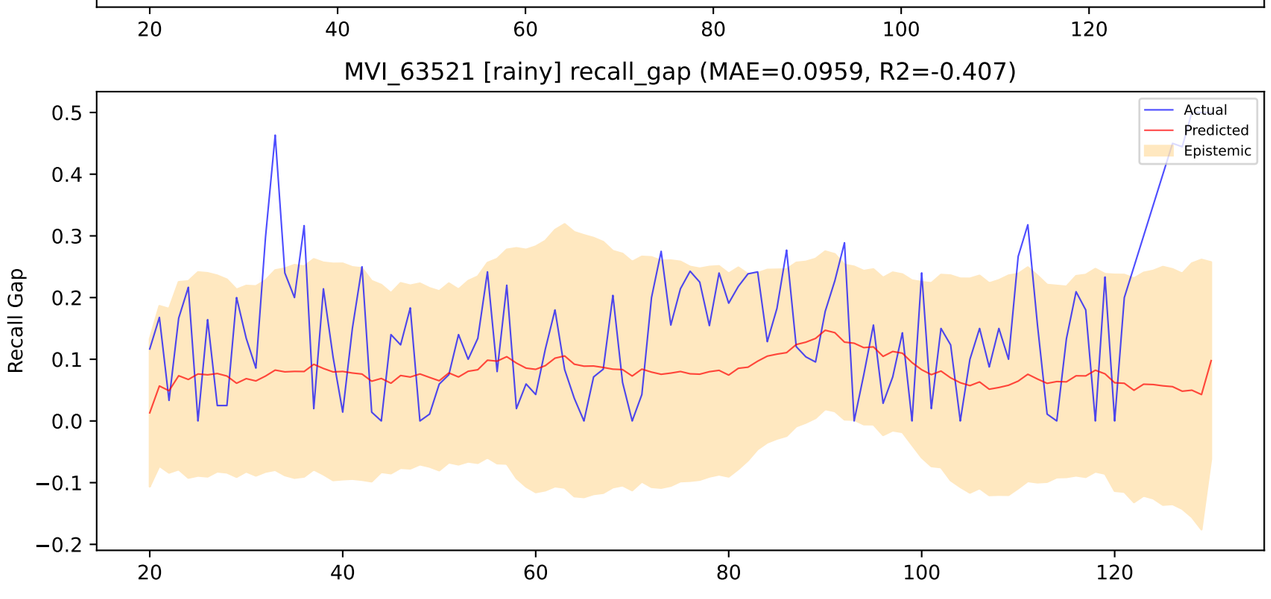
MVI_40732 [night] recall_gap (MAE=0.1458, R2=-0.180)



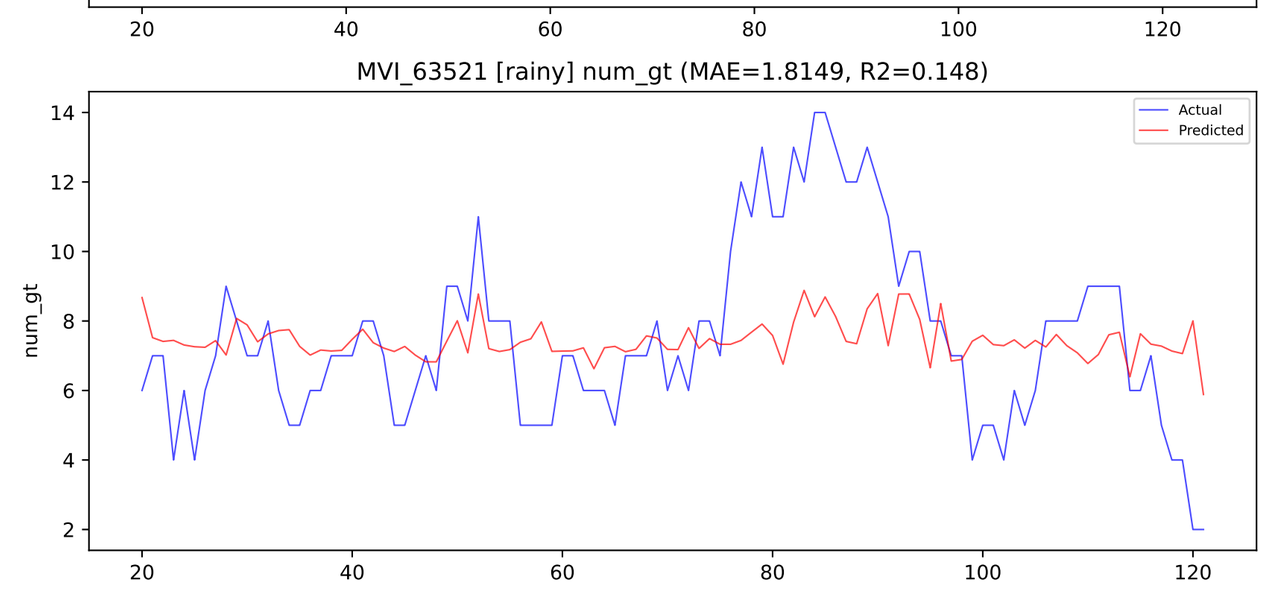
MVI_40732 [night] num_gt (MAE=2.1740, R2=-0.132)



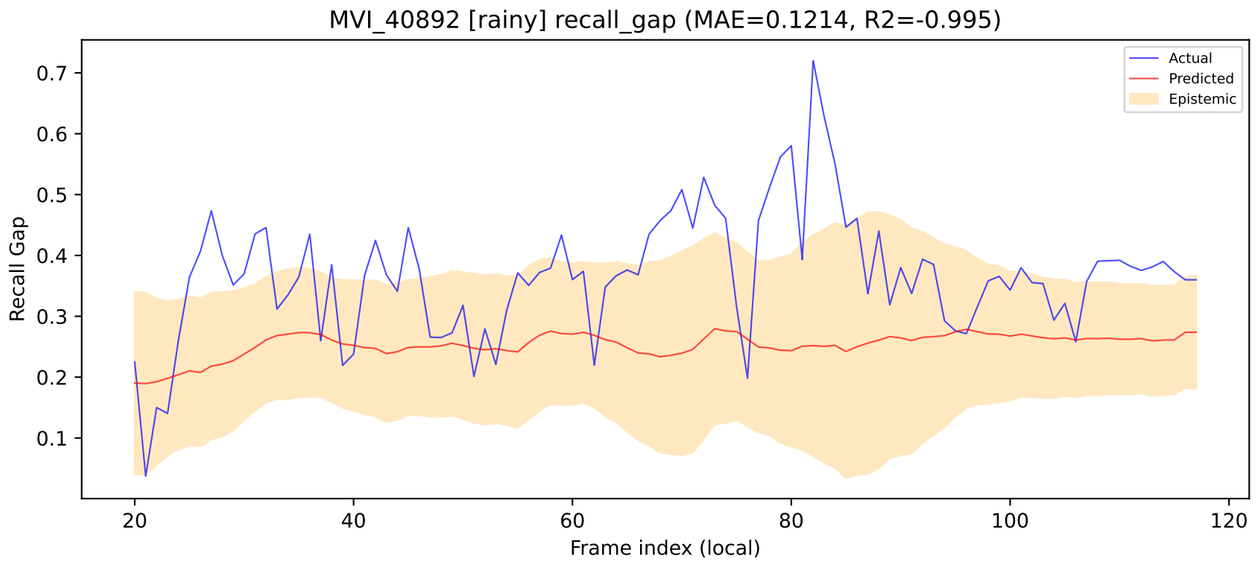
MVI_63521 [rainy] recall_gap (MAE=0.0959, R2=-0.407)



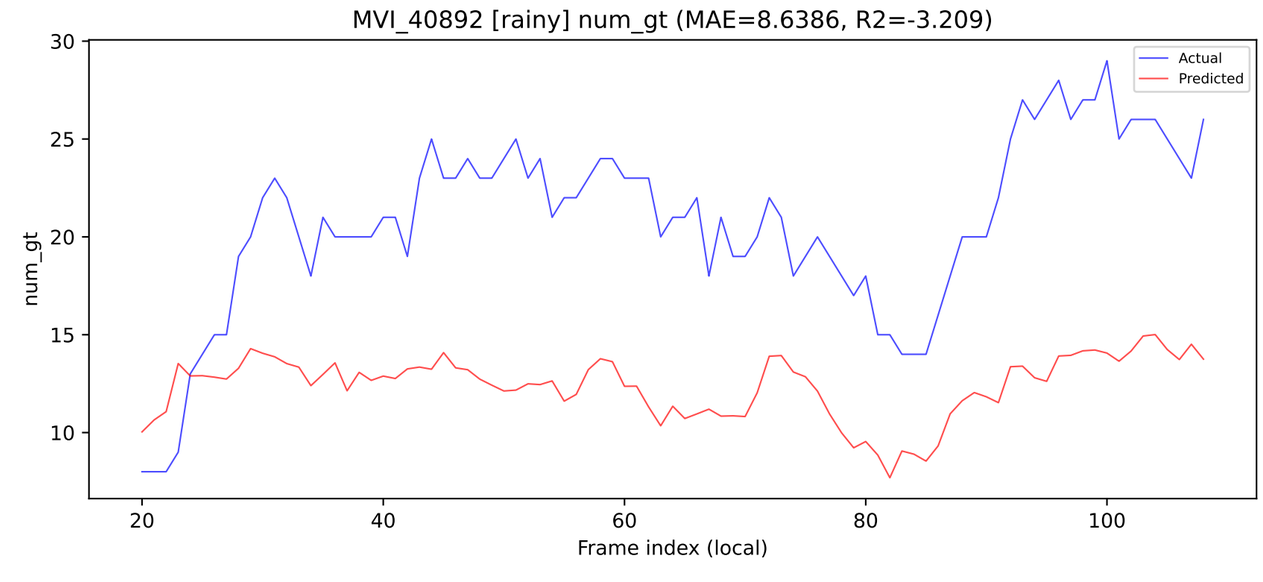
MVI_63521 [rainy] num_gt (MAE=1.8149, R2=0.148)



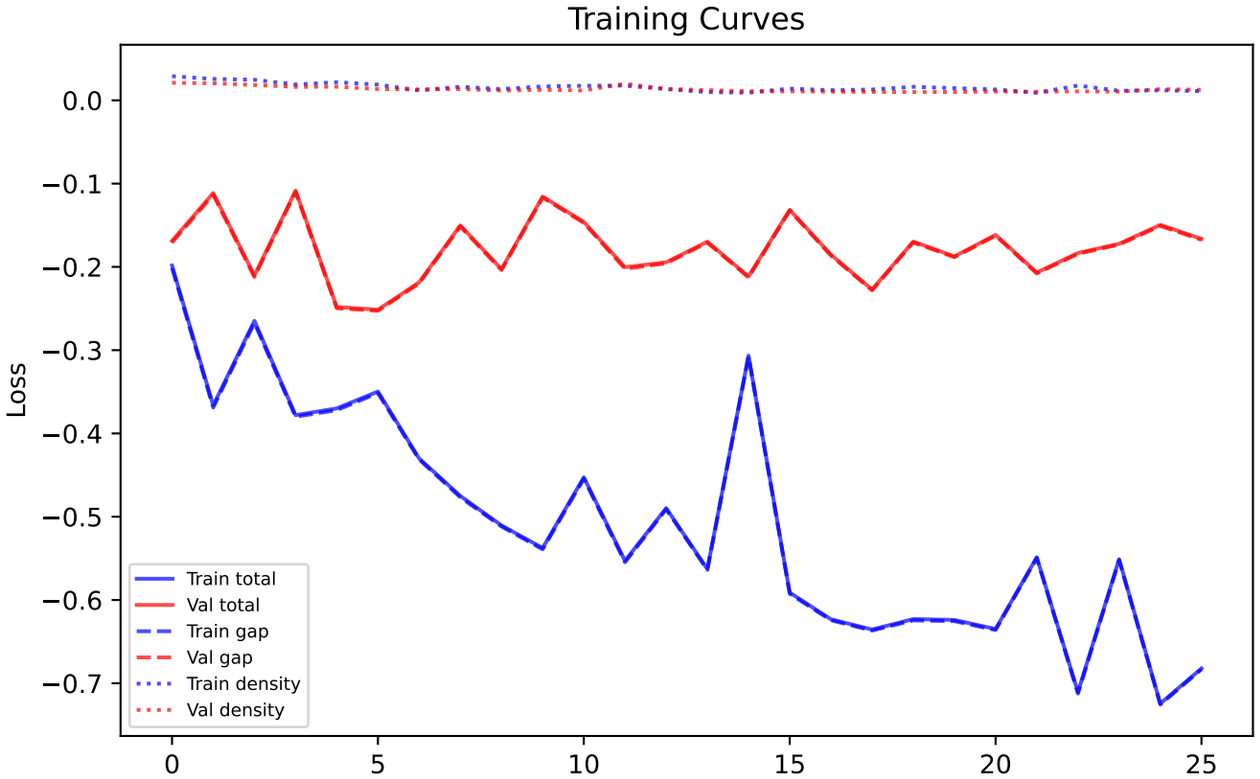
MVI_40892 [rainy] recall_gap (MAE=0.1214, R2=-0.995)

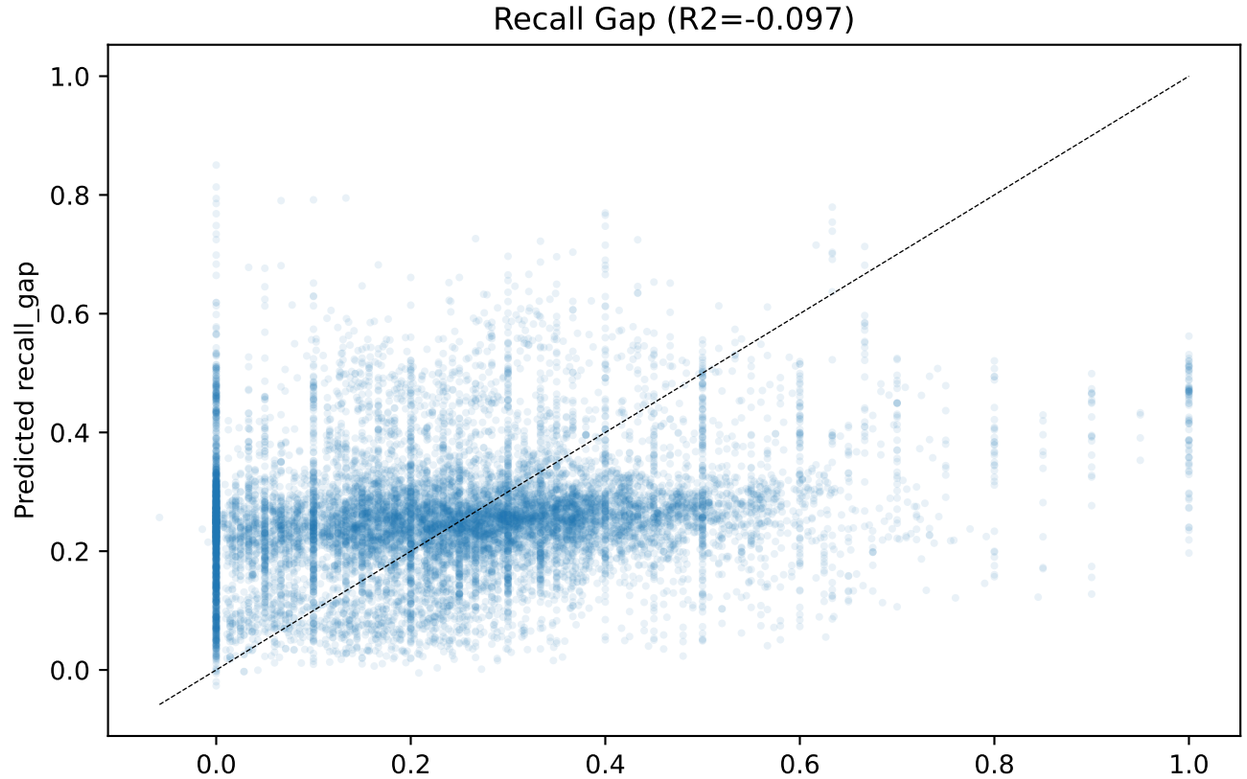


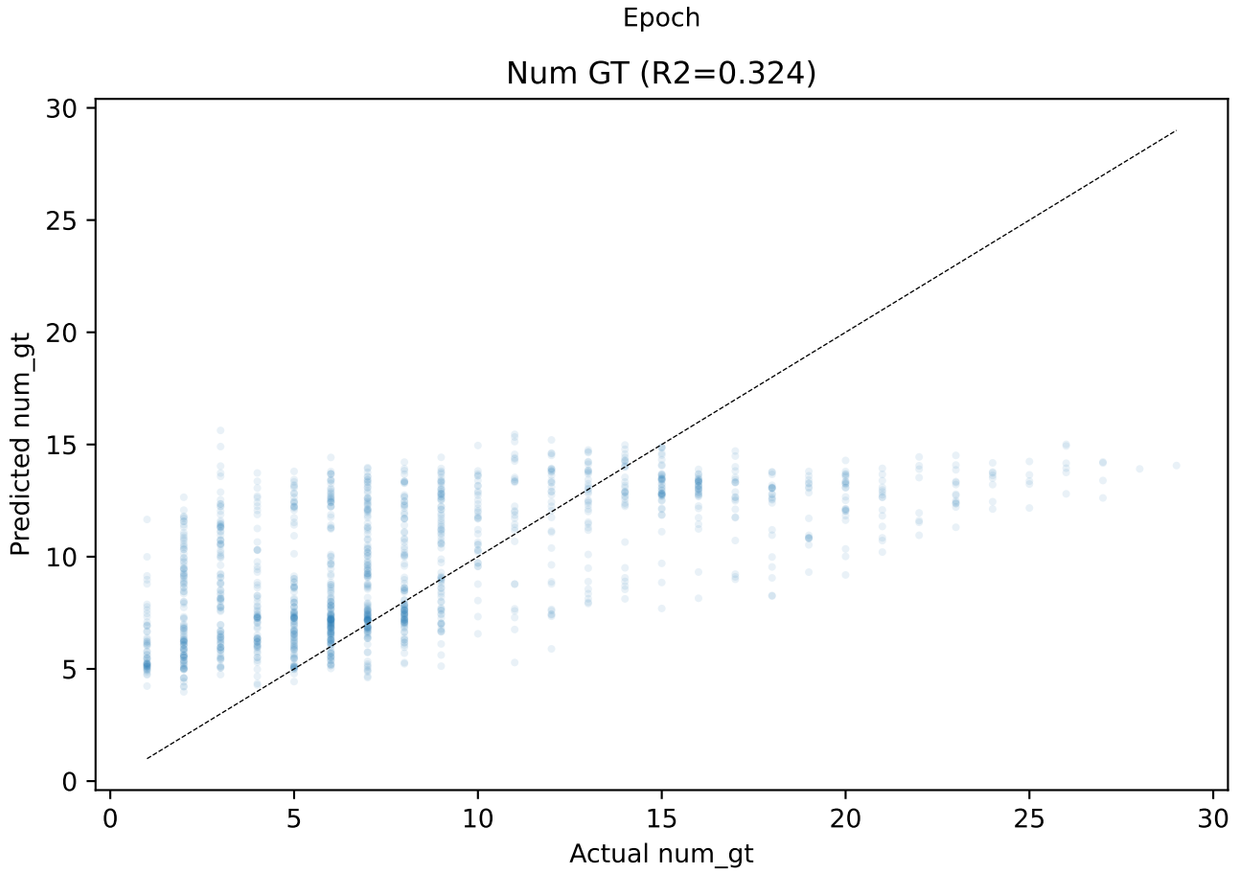
MVI_40892 [rainy] num_gt (MAE=8.6386, R2=-3.209)

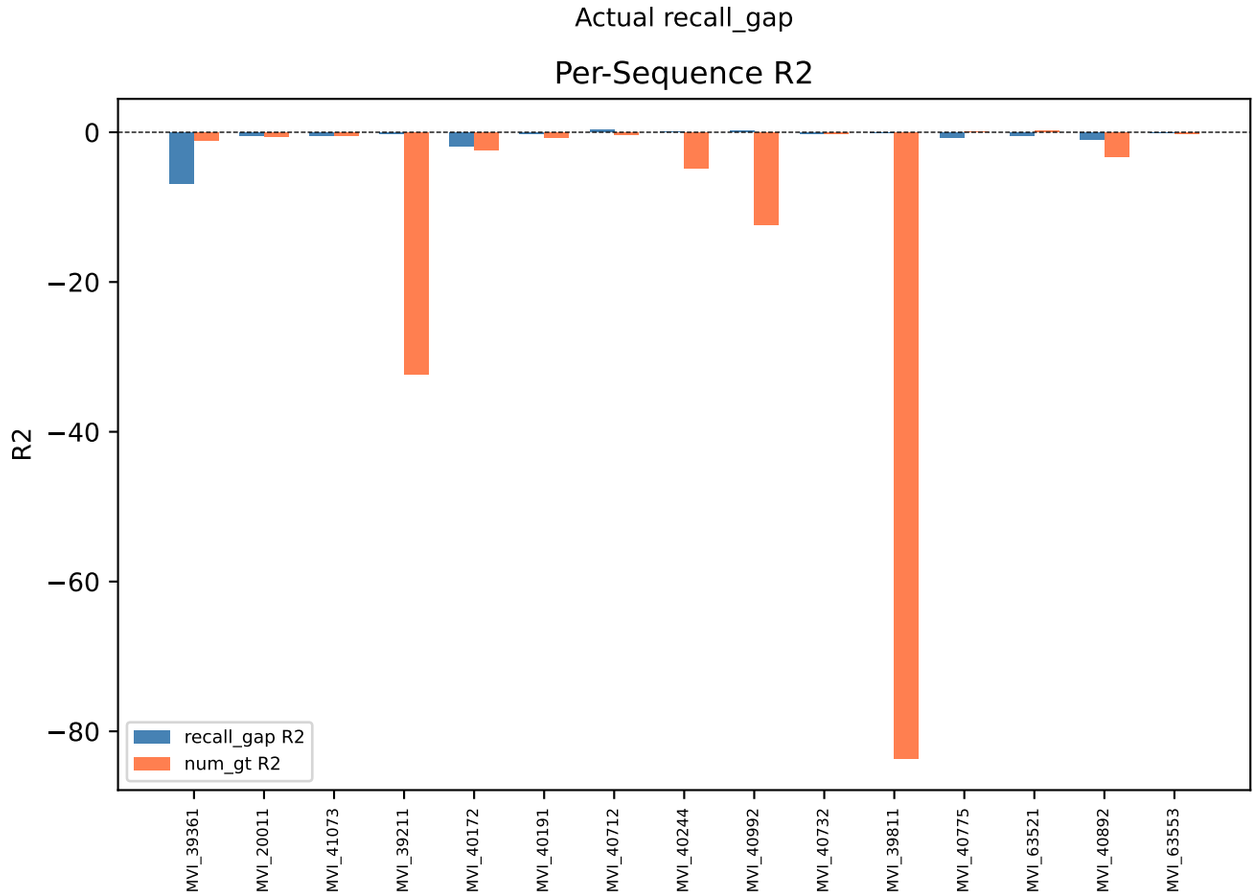


Page 2: Diagnostics









Page 3: Metrics & Comparison

LSTM_medium -- Validation Metrics (23 features: 14 image + 9 medium)

Sequence	Weather	Gap MAE	Gap R2	Gap rho	GT MAE	GT R2	GT rho
MVI_39361	sunny	0.2061	-6.8841	0.1626	5.0881	-1.0552	0.6957
MVI_20011	sunny	0.1231	-0.4529	-0.1441	2.5211	-0.5321	0.4506
MVI_41073	sunny	0.1456	-0.4607	-0.0037	1.6401	-0.4046	0.4986
MVI_39211	sunny	0.2523	-0.2402	0.2828	7.2508	-32.3598	0.2440
MVI_40172	cloudy	0.1640	-1.8414	0.0523	1.7125	-2.3190	0.7076
MVI_40191	cloudy	0.0742	-0.2042	-0.1103	3.4677	-0.7516	0.4947
MVI_40712	cloudy	0.1415	0.2589	0.7027	4.4694	-0.3469	0.3453
MVI_40244	cloudy	0.1041	0.0857	0.3388	5.1088	-4.8191	0.2436
MVI_40992	night	0.1748	0.1664	0.3872	3.6629	-12.3383	0.1225
MVI_40732	night	0.1458	-0.1795	0.1057	2.1740	-0.1317	0.0377
MVI_39811	night	0.3186	-0.0525	0.1092	3.9584	-83.6532	0.1390
MVI_40775	night	0.1077	-0.7318	0.1409	4.2268	0.0815	0.8647
MVI_63521	rainy	0.0959	-0.4069	0.1443	1.8149	0.1476	0.3330
MVI_40892	rainy	0.1214	-0.9949	0.3916	8.6386	-3.2089	0.6201
MVI_63553	rainy	0.0869	-0.0338	0.1042	1.6440	-0.2359	0.5087
OVERALL		0.1441	-0.0975	0.2151	3.7457	0.3240	0.6191
BASELINE	(mean)	0.1375	0.0000	nan	4.5528	0.0000	nan

Per-Weather + Cross-Experiment Comparison

Model / Weather	Gap R2	Gap MAE	n (windows)
sunny	-0.2390	0.19095	3000
cloudy	-0.2876	0.12360	4400
night	0.0251	0.16482	2820
rainy	0.1676	0.10223	2600
OVERALL (23 feat, medium)	-0.0975	0.14409	12820
Both models (32 feat)	0.1650	---	---
Image-only (14 feat)	-0.1150	---	---
UAVDT js10 (24 feat)	-0.2700	---	---